Answer 1

In a competitive market, equilibrium is reached when demand equals supply, that is D(pe) = S(pe). pe is the equilibrium price. If the system is not in equilibrium D(p) not equal to S(p)

A first-order differential equation has the general form

$dy/dx=f(x,y)$

If the right-hand side is a constant, say $f(x,y)=k$ then

$dy/dx=k$

Integrating,

$Dy = k\ dx$

$\int dy = \int k dx$

$y = kx+C$

Then, $y = kx+Cy = kx + Cy = kx+C$

where C is a constant of integration. This represents a straight line with constant slope k. The variable changes at a constant rate (steady trend). ($k>0$ measures how quickly price responds to excess demand.)

Example: supply - demand model of price adjustment. When the market is exactly in equilibrium, excess demand is zero, and thus

$dp/dt= 0$

The solution is $p = C$, meaning price remains constant, the system is at rest.

Economic intuition : If the right-hand side is constant, it means the variable changes at a steady, unchanging rate over time, independent of its current value. There is no feedback mechanism (no self-correction, no adjustment toward equilibrium). The variable simply grows or falls at a fixed slope forever.

Examples:

Steady inflation: If the price level P satisfies
$dP/dt= \pi$, where π is a constant inflation rate, then
$P(t) = \pi t + P_0$, Prices rise linearly over time at a constant rate.

Constant output growth:
If national income grows steadily by g units per year,
$dY/dt = g$, $Y(t) = gt + Y_0$,
Output increases at a constant pace.

Fixed rate of capital accumulation:
If investment adds a constant amount to capital each year,

$dK/dt = I_0$, capital stock rises linearly.

When the RHS is constant:

- The system does not adjust toward equilibrium.
- It describes uniform change like a car moving at constant speed.
- There is no stability analysis, since the rate of change never depends on deviations.

Answer 2

Part 1: Walrasian price adjustment differential equation

$$\dot{p}(t) = \theta(D(p(t)) - S(p(t))),$$

where $\theta > 0$is a constant, $D(p)$is demand and $S(p)$is supply.
Let p^*be an equilibrium price so that

$$D(p^*) = S(p^*).$$

deviation from equilibrium

$$x(t) \equiv p(t) - p^*.$$

So $p(t) = p^* + x(t)$. We will linearize around p^*.

First-order Taylor expansions of Dand S

Using the first-order Taylor (linear) approximation of Dand S about p^*. For a differentiable function f,

$$f(p) \approx f(p^*) + f'(p^*)\,(p - p^*)(\text{droping higher order terms}).$$

$$\begin{aligned} D(p) &\approx D(p^*) + D'(p^*)\,(p - p^*) = D^* + D'(p^*)\,x(t), \\ S(p) &\approx S(p^*) + S'(p^*)\,(p - p^*) = S^* + S'(p^*)\,x(t), \end{aligned}$$

where $D^* \equiv D(p^*)$and $S^* \equiv S(p^*)$. Because $D^* = S^*$(equilibrium), the constant terms cancel when we take $D - S$.

θ is the speed of adjustment parameter.

Substitute linearized expressions into the differential equation

$$\dot{p}(t) = \theta(D(p) - S(p))$$

becomes, using the approximations,

$$\dot{p}(t) \approx \theta[(D^* + D'(p^*)\,x(t)) - (S^* + S'(p^*)\,x(t))].$$

Simplifying the right hand side:

$$\dot{p}(t) \approx \theta[(D^* - S^*) + (D'(p^*) - S'(p^*))\, x(t)].$$

Because $D^* = S^*$, the constant term vanishes:

$$\dot{p}(t) \approx \theta\, (D'(p^*) - S'(p^*))\, x(t).$$

$x(t) = p(t) - p^*$. Differentiating $x(t)$:

$$\dot{x}(t) = \dot{p}(t) - \dot{p} * = \dot{p}(t).$$

So we can rewrite the equation entirely in terms of $x(t)$:

$$\dot{x}(t) = \theta(D'(p^*) - S'(p^*))\, x(t).$$

Putting it in standard linear ODE form

Let

$$\lambda \equiv \theta(D'(p^*) - S'(p^*)).$$

Then the linear ODE is

$$\dot{x}(t) = \lambda\, x(t).$$

This is the first-order linear homogeneous ODE with constant coefficient.

Solving the ODE (separation of variables)

solving $\dot{x} = \lambda x$by separation of variables:

$\frac{dx}{dt} = \lambda x.$

Separate: $\frac{dx}{x} = \lambda\, dt$.(Assume $x \neq 0$; the $x = 0$ solution is the equilibrium.)

Integrating both sides:

$$\int \frac{dx}{x} = \int \lambda\, dt.$$

Evaluating integrals:

$$\ln |\, x \,| = \lambda t + C,$$

where C is an integration constant.
Exponentiating:

$$|\, x \,| = e^{\lambda t + C} = e^{C} e^{\lambda t}.$$

Let $A \equiv \pm e^{C}$(absorbing sign into the constant) so

$$x(t) = Ae^{\lambda t}.$$

Using initial condition to pin down A. Suppose $p(0)$is given, so $x(0) = p(0) - p^*$. Then

$$x(0) = Ae^{\lambda \cdot 0} = A \text{ So, } A = p(0) - p^*.$$

The solution for $p(t)$

$p(t) = p^* + x(t)$. Hence

$$\boxed{p(t) = p^* + (p(0) - p^*)\, e^{\lambda t}}$$

with $\lambda = \theta(D'(p^*) - S'(p^*))$. Writing λ explicitly,

$$\boxed{p(t) = p^* + (p(0) - p^*) \exp(\theta(D'(p^*) - S'(p^*))t)\,.}$$

Stability interpretation

Stability depends on the sign of λ.

- If $\lambda < 0$, then $e^{\lambda t} \to 0$ as t tends to ∞. So $p(t)$ tends to p^*: the equilibrium is locally asymptotically stable.
- If $\lambda > 0$, then $e^{\lambda t} \to \infty$and deviations grow: equilibrium is unstable.
- If $\lambda = 0$, the linear term vanishes; the linear approximation is inconclusive and higher order terms determine dynamics.

Given standard economic shapes:

- $D'(p^*) < 0$(demand falls with price),
- $S'(p^*) > 0$(supply rises with price),
- and $\theta > 0$,

we have $D'(p^*) - S'(p^*) < 0$, hence $\lambda < 0$and the equilibrium is (typically) stable.

Economic intuition:
In real markets, this mechanism captures how prices act as signals:

- When there's excess demand, prices rise, encourages producers to supply more, consumers to demand less, restores balance.
- When there's excess supply, prices fall, discourages production, encourages consumption, again restores balance.

This dynamic of prices adjusting until markets clear is the backbone of Walrasian general equilibrium theory and explains why under normal conditions, markets tend to stabilize rather than explode.

The Walrasian price adjustment mechanism describes how prices in a market change in response to imbalances between demand and supply. When demand exceeds supply, prices

rise, and when supply exceeds demand, prices fall, moving the market toward equilibrium. Near the equilibrium price, small deviations are corrected through these adjustments, depending on how responsive buyers and sellers are to price changes. If demand falls when price rises and supply increases when price rises, what typically happens in real markets, then prices naturally move back to equilibrium, creating stability. This means that when there is a shortage, prices increase to reduce demand and boost supply, and when there is a surplus, prices decrease to encourage demand and reduce supply. However, if these relationships were reversed, for example, if higher prices made people want to buy more or producers supply less, then prices would move away from equilibrium, making the market unstable. In essence, this mechanism captures how prices act as signals that coordinate the behavior of consumers and producers, ensuring that markets tend to balance themselves over time rather than spiral out of control.

Part 2: STABILITY PROPERTIES OF BOTH CASES

Stability Properties of the Walrasian Price Adjustment Model

1. Stable Case (Normal Market Conditions)

- Condition: Demand decreases with price $(D'(p^*) < 0)$, and supply increases with price $(S'(p^*) > 0)$.
- Mechanism:
  If price is below equilibrium, there is excess demand, prices rise.
  If price is above equilibrium, there is excess supply, prices fall.
- Outcome: Deviations from equilibrium are corrected over time. Prices converge back to p^*.
- Economic Intuition: The market is self-correcting. Prices act as signals that balance demand and supply.
- Stability: Locally asymptotically stable — small disturbances die out over time.

2. Unstable Case (Abnormal Market Conditions)

- Condition: Demand increases with price $(D'(p^*) > 0)$ or supply decreases with price $(S'(p^*) < 0)$.
- Mechanism:
  If price rises slightly, excess demand grows, prices rise further.
  If price falls slightly, excess supply increases, prices fall further.
- Outcome: Deviations from equilibrium amplify instead of shrinking. Prices move away from p^*.
- Economic Intuition: The market is self-reinforcing rather than self-correcting. Prices fail to stabilize.
- Stability: Unstable equilibrium — small shocks cause diverging price movements.

Stable equilibrium: $D'(p^*) - S'(p^*) < 0$, prices return to equilibrium.

Unstable equilibrium: $D'(p^*) - S'(p^*) > 0$, prices diverge from equilibrium.

The stability of the Walrasian price adjustment model depends entirely on how prices respond to excess demand or supply, which in turn depends on the slopes of the demand and supply functions. When the slope of demand is negative (meaning consumers buy less as prices rise)

and the slope of supply is positive (meaning producers supply more as prices rise), the adjustment process is stable. In this case, if the price is below the equilibrium, excess demand pushes it upward, and if the price is above equilibrium, excess supply pushes it downward, causing prices to return smoothly to the equilibrium level over time. This is the situation where markets self-correct.

In contrast, if the opposite occurs, demand increases with price or supply decreases with price, the adjustment process becomes unstable. Here, a small deviation from equilibrium sets off forces that drive prices even further away: if price rises slightly, excess demand increases, causing further upward pressure; if price falls slightly, excess supply grows, pushing the price down even more. This leads to diverging price movements rather than convergence. In short, the system is stable when normal downward-sloping demand and upward-sloping supply ensure corrective price movements, and unstable when these slopes are reversed, causing prices to move explosively away from equilibrium.

Part 3:

From part 1, we have:

$$p(t) = p^* + (p(0) - p^*)e^{\lambda t}, \text{with } \lambda = \theta(D'(p^*) - S'(p^*))$$

- Stability condition: $\lambda < 0 \implies D'(p^*) - S'(p^*) < 0$
- price converges smoothly and monotonically to the equilibrium p^*if the usual economic condition holds: $D' < 0, S' > 0, \theta > 0$.

The discrete-time Cobweb model

The Cobweb model is:

$$p_{t+1} = f(p_t) = S^{-1}(D(p_t))$$

- deviation from equilibrium: $x_t = p_t - p^*$.
- Linearizing $f(p_t)$ around p^*:

$$x_{t+1} \approx f'(p^*)x_t$$

- Here, $f'(p^*) = \frac{D'(p^*)}{S'(p^*)}$, assuming differentiable supply and demand.

Stability condition for the Cobweb model

- The linearized solution:

$$x_t = (f'(p^*))^t x_0$$

- Stability requires:

$$| f'(p^*) | < 1 \implies | \frac{D'(p^*)}{S'(p^*)} | < 1$$

1. $0 < f'(p^*) < 1 \rightarrow$ monotone convergence
2. $-1 < f'(p^*) < 0 \rightarrow$ oscillatory convergence
3. $| f'(p^*) | > 1 \rightarrow$ divergence (unstable)

Comparing continuous vs discrete-time dynamics

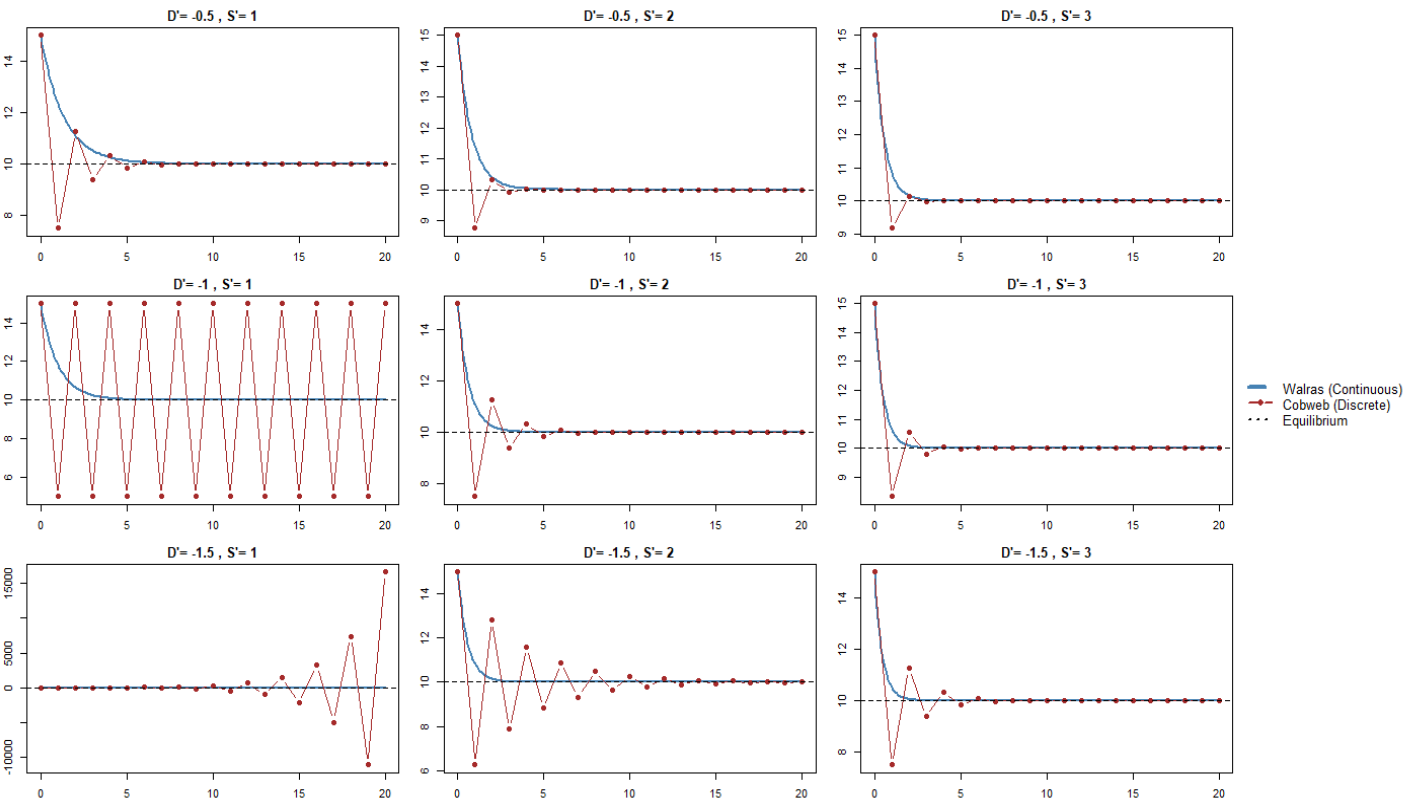
	Continuous-time Walrasian	Discrete-time Cobweb
Equation	$\dot{x} = \lambda x, \lambda = \theta(D' - S')$	$x_{t+1} = f'(p^*)x_t, f'(p^*) = D'/S'$
Stability condition	$\lambda < 0 \implies D' - S' < 0$	$| f'(p^*) | < 1 \implies \left| \frac{D'(p^*)}{S'(p^*)} \right| < 1$ $0 < f'(p^*) < 1 \rightarrow$ monotone convergence $-1 < f'(p^*) < 0 \rightarrow$ oscillatory convergence
Convergence pattern	Monotone, exponential decay	Monotone if $0 < f' < 1$, oscillatory if $-1 < f' < 0$
Divergence	$\lambda > 0$: exponential growth	$| f'(p^*) | > 1 \rightarrow$ divergence (unstable)
Adjustment style	Continuous, gradual	Stepwise, per period, may overshoot

1. Continuous-time system is inherently smoother, adjustments are gradual, and standard demand/supply slopes almost always guarantee stability.
2. Discrete-time Cobweb model can oscillate or even diverge, depending on the relative slopes of demand and supply ($| D'/S' |$).
3. The risk of price oscillations or instability is higher in discrete-time because the system updates in jumps, not continuously.

Economic interpretation

- Walrasian adjustment: Markets with continuous, small price adjustments tend to converge steadily to equilibrium.
- Cobweb dynamics: Markets with discrete, period-by-period adjustments (e.g., agricultural markets with time lags) may exhibit cycles or overshooting, even if the continuous-time counterpart is stable.

Answer 3: Simulation



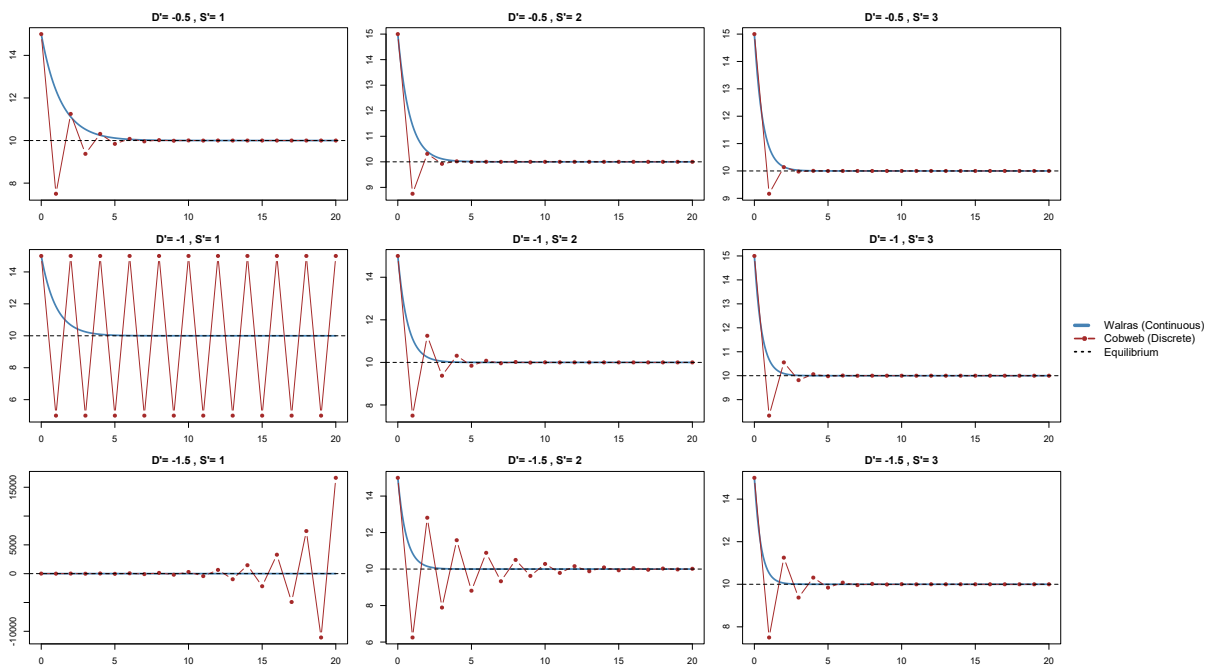
The simulation results show that when demand slopes downward and supply slopes upward (the usual case), both the Walrasian and Cobweb models converge smoothly to the equilibrium price. However, as the absolute value of the demand slope increases or the supply slope becomes steeper, the Cobweb model may start to oscillate or even diverge, while the continuous-time Walrasian model remains stable and adjusts gradually. This illustrates how discrete, period-by-period price adjustments can create cyclical or unstable outcomes, whereas continuous adjustments tend to restore equilibrium more smoothly.

The Walrasian continuous-time price adjustment mechanism can diverge only if the effective adjustment parameter $\lambda = \theta(D'(p^*) - S'(p^*))$is positive, meaning that deviations from equilibrium grow exponentially over time instead of shrinking. For this to happen, either the demand curve must slope upwards ($D'(p^*) > 0$), so that higher prices induce higher quantity demanded, or the supply curve must slope downwards ($S'(p^*) < 0$), so that higher prices reduce quantity supplied. Both of these conditions are contrary to standard economic intuition, where demand normally falls with price ($D' < 0$) and supply rises with price ($S' > 0$). If $\lambda > 0$, the system becomes unstable and any initial deviation from equilibrium is amplified continuously, causing the price to diverge away from p^*. In typical economic scenarios, however, $\lambda < 0$and the Walrasian system remains stable.

IDAS Assignment_Oct

Michelle

2025-10-16

```r
p_star <- 10          # equilibrium price
p0 <- 15              # initial price
theta <- 0.5          # speed of adjustment for Walras
t_max <- 20           # max time for continuous simulation
t_discrete <- 20      # number of discrete periods

# Parametric variation: slopes of demand and supply
D_primes <- c(-0.5, -1, -1.5)  # demand slope (negative)
S_primes <- c(1, 2, 3)         # supply slope (positive)

# Time vector for continuous-time
t_cont <- seq(0, t_max, by = 0.1)

# ---- Layout: 3x3 grid for plots + 1 column for legend ----
layout_matrix <- cbind(
  matrix(1:(length(D_primes) * length(S_primes)),
         nrow = length(D_primes), byrow = TRUE),
  rep(length(D_primes)*length(S_primes) + 1, length(D_primes))
)
layout(layout_matrix, widths = c(rep(3, length(S_primes)), 1.5))

par(mar = c(2.5,2.5,2,1), oma = c(0,0,0,0))


# ---- Loop for plots ----
for(Dp in D_primes){
  for(Sp in S_primes){

    # Continuous-time Walras
    lambda <- theta * (Dp - Sp)
    p_walras <- p_star + (p0 - p_star) * exp(lambda * t_cont)

    # Discrete-time Cobweb
    f_prime <- Dp / Sp
    p_cobweb <- numeric(t_discrete + 1)
    p_cobweb[1] <- p0
    for(t in 1:t_discrete){
      p_cobweb[t+1] <- p_star + f_prime * (p_cobweb[t] - p_star)
    }

    # Plot both
    plot(t_cont, p_walras, type="l", col="steelblue", lwd=2,
```

```r
        ylim = range(c(p_walras, p_cobweb)),
        xlab="", ylab="", main = paste("D'=",Dp,", S'=",Sp))
    points(0:t_discrete, p_cobweb, type="b", col="brown", pch=16)
    abline(h=p_star, lty=2)
  }
}

# ---- Legend panel ----
par(mar = c(0,0,0,0))
plot.new()
legend("left",
       legend = c("Walras (Continuous)", "Cobweb (Discrete)", "Equilibrium"),
       col = c("steelblue", "brown", "black"),
       lty = c(2, 2, 3),
       lwd = c(4, 2, 2),
       pch = c(NA, 16, NA),
       cex = 1.2,
       bty = "n")
```